PASSAGE SUMMARIZATION WITH RECURRENT MODELS FOR AUDIO – SHEET MUSIC RETRIEVAL

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ABSTRACT

Many applications of cross-modal music retrieval are related to connecting sheet music images to audio recordings. A typical and recent approach to this is to learn, via deep neural networks, a joint embedding space that correlates short fixed-size snippets of audio and sheet music by means of an appropriate similarity structure. However, two challenges that arise out of this strategy are the requirement of strongly aligned data to train the networks, and the inherent discrepancies of musical content between audio and sheet music snippets caused by local and global tempo differences. In this paper, we address these two shortcomings by designing a cross-modal recurrent network that learns joint embeddings that can summarize longer passages of corresponding audio and sheet music. The benefits of our method are that it only requires weakly aligned audio – sheet music pairs, as well as that the recurrent network handles the non-linearities caused by tempo variations between audio and sheet music. We conduct a number of experiments on synthetic and real piano data and scores, showing that our proposed recurrent method leads to more accurate retrieval in all possible configurations.

1. INTRODUCTION

The abundance of music-related content in various digital formats, including studio and live audio recordings, scanned sheet music, and metadata, among others, calls for efficient technologies for cross-linking between documents of different modalities. In this work, we explore a cross-modal task referred to as audio – sheet music passage retrieval. We define it as follows: given an audio fragment as a query, search within an image database and retrieve the corresponding sheet music passage; or vice versa, find the appropriate recording fragment given a query in the form of some snippet of (scanned) sheet music.

A fundamental step in audio–sheet music retrieval concerns defining a suitable shared representation that permits the comparison between items of different modalities in a convenient and effective way. The conventional approaches for linking audio recordings to their respective printed scores are based on handcrafted mid-level representations [1, 2]. These are usually pitch-class profiles, like chroma-based features [3,4], symbolic fingerprints [5], or the bootleg score [6, 7], which is a coarse mid-level codification of the main note-heads in a sheet music image. However extracting such representations requires a series of pre-processing stages that are prone to errors, for example optical music recognition on the sheet music side [8–10], and automatic music transcription on the audio part [11–13].

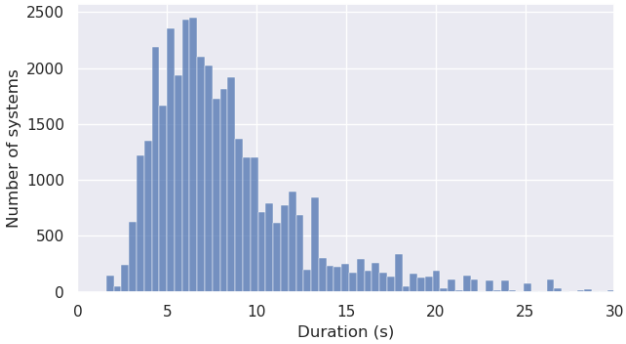


Figure 1: Distribution of system durations in around 40,000 examples from the MSMD. More than 25% of the passages are longer than ten seconds.

A promising approach [14, 15] has been proposed to eliminate these problematic pre-processing steps by learning a shared low-dimensional embedding space directly from audio recordings and printed scores. This is achieved by optimizing a cross-modal convolutional network (CNN) to project short snippets of audio and sheet music onto a latent space, in which the cosine distances between semantically related snippets are minimized, whereas non-related items of either modality are projected far from each other. Then the retrieval procedure is reduced to simple nearest-neighbour search in the shared embedding space, which is a simple and fast algorithm.

A first limitation of this strategy relates to its supervised nature: it requires strongly-aligned data in order to generate matching audio–sheet snippet pairs for training, which means fine-grained mappings between note onsets and corresponding note positions in the score. Obtaining such annotations is tedious and time-consuming, and also

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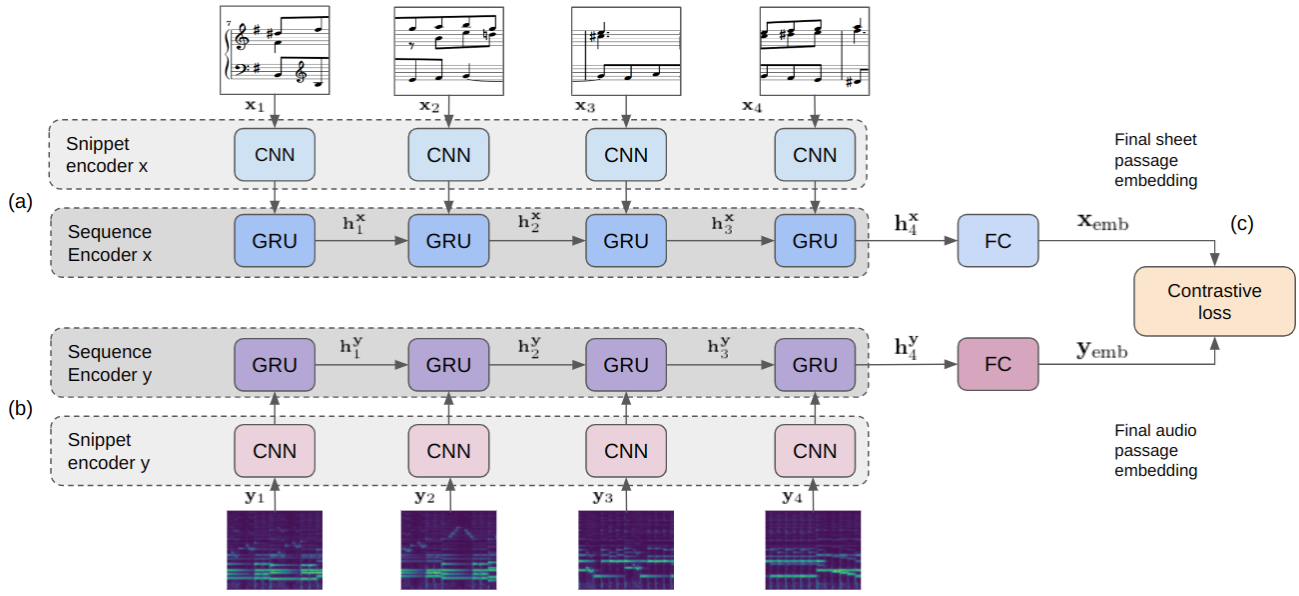


Figure 2: Diagram of the proposed network. Two independent pathways are trained to encode sheet music (a) and audio (b) passages by minimizing a contrastive loss function (c).

requires specialized annotators with musical training. As a result, embedding learning approaches have been trained with synthetic data, in which recordings, sheet music images, and their respective alignments are rendered from symbolic scores. This leads to poor generalization in scenarios with real music data, as shown in [16].

Moreover, the snippets in both modalities have to be fixed in size, meaning that the amount of actual musical content in the fragments can vary considerably depending on note durations and the tempo in which the piece is played. For example, a sheet excerpt with longer notes played slowly would correspond to a considerably larger duration in audio than one with short notes and a faster tempo. This leads to generalization problems caused by differences between what the model sees during training and test time; [17] attempted to address this limitation by introducing a soft-attention mechanism to the network.

In this paper we address the two aforementioned limitations by proposing a recurrent cross-modal network that learns compact, fixed-size representations from longer variable-length fragments of audio and sheet music. By removing the fixed-size fragment constraint, we can adjust the lengths of fragments during training so that cross-modal pairs can span the same music content, leading to a more robust representation. Moreover, by operating with longer music passages, it is possible to rely solely on weakly-annotated data for training, since we now require only the starting and ending positions of longer-context music fragments within music documents, in order to extract audio–sheet passages to prepare a train set. This is a remarkable advantage compared for example to other approaches based on [14], where fine-detailed alignments are indispensable to generate short audio–sheet snippet pairs.

The rest of the paper is structured as follows. In Section 2 we describe the model proposed to learn joint representations from cross-modal passages. Section 3 presents a series of experiments on artificial and real data and Section 4 summarizes and concludes the work.

2. AUDIO–SHEET PASSAGE RETRIEVAL

For the purposes of this paper, and in order to be able to use our annotated corpora for the experiments, we define a "passage" as the musical content corresponding to one line of sheet music (also known as a "system"). System-level annotation of scores are much easier to come by than note-precise score-recording alignments, making it relatively easy to compile large collections of training data for our approach. Our definition of passages resembles that of "musical themes", which has been used under a cross-modal retrieval scenario with symbolic queries in a number of previous works [18, 19]. To illustrate the temporal discrepancies between passages, we show in Figure 1 the distribution of time duration of the systems from all pieces of the MSMD dataset [14] (later we will elaborate more on this database). In this dataset, we observe that systems can cover from less than five to more than 25 seconds of musical audio.

This important temporal aspect motivates us to propose the network depicted in Figure 2 to learn a common latent representation from pairs of audio–sheet passages. The architecture has two independent recurrent-convolutional pathways, which are responsible for encoding sheet music (Figure 2a) and audio (Figure 2b) passages. The key component of this approach is the introduction of two recurrent layers that, inspired by traditional sequence-to-sequence models [22], are trained to summarize a variable-length sequences into context vectors, that we conveniently refer to as embedding vectors.

Defining a pair of corresponding passages in the form of image (sheet music) and log-magnitude spectro-

Audio CNN encoder input: 92×20	**Sheet-Image CNN encoder** input: 160×180
2x Conv(3, pad-1)-24 - BN	2x Conv(3, pad-1)-24 - BN
MaxPooling(2)	MaxPooling(2)
2x Conv(3, pad-1)-48 - BN	2x Conv(3, pad-1)-48 - BN
MaxPooling(2)	MaxPooling(2)
2x Conv(3, pad-1)-96 - BN	2x Conv(3, pad-1)-96 - BN
MaxPooling(2)	MaxPooling(2)
2x Conv(3, pad-1)-96 - BN	2x Conv(3, pad-1)-96 - BN
MaxPooling(2)	MaxPooling(2)
Conv(1, pad-0)-32 - BN	Conv(1, pad-0)-32 - BN
FC(32)	FC(32)

Table 1: Overview of the two convolutinal encoders. Each side is responsible for their respective modality. Conv(3, pad-1)-24: 3×3 convolution, 24 feature maps and zero-padding of 1. BN: Batch normalization [20]. We use ELU activation functions [21] after all convolutional and fully-connected layers.

gram (audio) as $\mathbf{X}$ and $\mathbf{Y}$, respectively, two sequences $(\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_N})$ and $(\mathbf{y_1}, \mathbf{y_2}, \ldots, \mathbf{y_M})$ are generated by sequentially cutting out short snippets from $\mathbf{X}$ and $\mathbf{Y}$. The shapes of the short sheet and audio snippets are respectively 160×180 (pixels) [1] and 92×20 (frequency bins $\times$ frames), which corresponds to one second of audio. After that, each individual snippet is encoded by a VGG-style CNN [23] into a 32-dimensional vector, as shown in Figure 2, generating two sequences of encoded snippets, one for the audio passage, and the other for the sheet passage (note that each modality has its own dedicated CNN encoder). The architecture of the CNN encoders are detailed in Table 1.

Then each sequence is fed to a recurrent layer in order to learn the spatial and temporal relations between subsequent snippets, which are inherent in music. After experimenting with two typical simple recurrent layers, namely long short-term memory cells (LSTM) [24] and gated recurrent units (GRU) [25], we observed on average better results with GRUs, and we decided for the latter for our architecture. Each of the two GRUs is designed with 128 hidden units, where the hidden state of each GRU after the last step is the context vector that summarizes the passages. Finally a fully connected layer (FC) is applied over each context vector, in order to encode the final passage embeddings $(\mathbf{x}_{\text{emb}}, \mathbf{y}_{\text{emb}})$ with the desired dimension.

During training, a triplet (contrastive) loss function [26] is used to minimize the distances between embeddings from corresponding passages of audio and sheet music and maximize the distance between non-corresponding ones. Defining $\mathrm{d}(\cdot)$ as the cosine distance, the loss function is given by:

$$\mathcal{L} = \sum_{k=1}^{K} \max\Big\{0, \alpha + \mathrm{d}\Big(\mathbf{x}_{\text{emb}}, \mathbf{y}_{\text{emb}}\Big) - \mathrm{d}\Big(\mathbf{x}_{\text{emb}}, \mathbf{y}^{k}_{\text{emb}}\Big)\Big\}, \tag{1}$$

where $\mathbf{y}^{k}_{\text{emb}}$ for $k \in 1, 2, \ldots, K$ are contrastive (negative) examples from K non-matching passages in the same training mini-batch. This contrastive loss is applied to all $(\mathbf{x}_{\text{emb}}, \mathbf{y}_{\text{emb}})$ pairs within each mini-batch iteration. The margin parameter $\alpha \in \mathbb{R}_{+}$, in combination with the $\max\{\cdot\}$ function, penalizes matching snippets that were poorly embedded.

For the sake of simplicity, we leave the remaining details concerning the design of the networks, such as learning hyper-parameters, to our repository where our method will be made publicly available, [2] as well as the trained models derived in this work.

3. EXPERIMENTS

In this section we conduct experiments on different audio–sheet music scenarios. We first elaborate on the main dataset used for training and evaluation and define the steps of the passage retrieval task. Then we select four experiment setups and present the results.

We train our models with the Multi-Modal Sheet Music Dataset (MSMD) [14], which is a collection of classical piano pieces with multifaceted data, including score sheets (PDF) engraved via Lilypond [3] and corresponding audio recordings rendered from MIDI with several types of piano soundfonts. With over 400 pieces from over 50 composers, including Bach, Beethoven and Schubert, and covering more than 15 hours of audio, the MSMD has audio–sheet music alignments which allow us to obtain corresponding cross-modal pairs of musical passages. From the MSMD we were able to derive roughly 5,000 audio–sheet passages for training, which is scaled up to around 40,000 different pairs after data augmentation: audios are re-rendered with different soundfonts and have their tempo changed between 90% and 110%. Then we generate a test set of 534 pairs from a separate set of music pieces, that were rendered with a soundfont that was not seen during training. Later, in 3.2, we will also consider real scanned scores and real audio recordings.

To perform cross-modal passage retrieval, we first embed all audio–sheet pairs in the shared space using our trained model depicted in Figure 2. Then the retrieval is conducted by using the cosine distance and nearest-neighbor search within the space. For example, in case of using an audio passage as a query to find the appropriate sheet music fragment, the pairwise cosine distances between the query embedding and all the sheet music passage embeddings are computed. Finally, the retrieval results are obtained by means of a ranked list through sorting the distances in ascending order.

As for evaluation metrics, we look at the *Recall@k* (R@k), *Mean Reciprocal Rank* (MRR) and the *Median Rank* (MR). The R@k measures the ratio of queries which were correctly retrieved within the top k results. The MRR is defined as the average value of the reciprocal rank over all queries. MR is the median position of the correct match in the ranked list.

[1] In our approach, all sheet music pages are initially re-scaled to a 1181×835 resolution

[2] `https://github.com/luisfvc/lcasr`

[3] `http://www.lilypond.org`

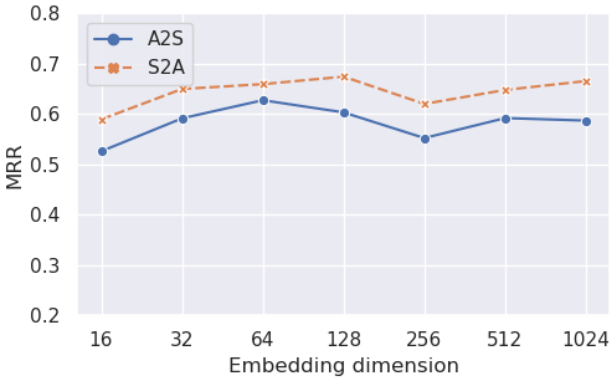


Figure 3: Mean Reciprocal Rank (MRR) for different embedding dimensions, evaluated in both search directions.

3.1 Experiment 1: Embedding dimension

In the first round of experiments, we investigate the effect of the final embedding dimension in the retrieval task. We consider the values in $\{16, 32, 64, 128, 256, 512, 1024\}$ and train the model of Figure 2 with the same hyperparameters. Then we perform the retrieval task in both search directions: audio-to-sheet music (A2S) and sheet music-to-audio (S2A).

Figure 3 presents the MRR of the snippet retrieval results evaluated on the 534 audio–sheet music passage pairs of the MSMD testset. A first and straightforward observation is that in all cases the S2A direction indicates better retrieval quality. We observe the performance increasing together with the embedding dimensionality until it stagnates at 64-D, and the MRR does not improve on average for higher-dimensional embeddings. For this reason, we select the model that generates 64-dimensional embeddings as the best one, which will be evaluated more thoroughly in the next experiments.

3.2 Experiment 2: Real data and improved models

In this section, we conduct an extensive series of experiments comparing our proposed recurrent network and some improved models thereof with baseline methods, and extend the evaluation to real-world piano data.

Given that our training data are entirely synthetic, we wish to investigate the generalization of our models from synthetic to real data. To this end, we evaluate on three datasets: on a (1) fully artificial one, and on datasets consisting (2) partially and (3) entirely of real data. For (1) we use the test split of MSMD and for (2) and (3) we combine the Zeilinger and Magaloff Corpora [27] with a collection of commercial recordings and scanned scores that we have access to. These data account for more than a thousand pages of sheet music scans with mappings to both MIDI files and over 20 hours of classical piano recordings. Then, besides the MSMD (I), we define two additional evaluation sets: (II) *RealScores_Synth*: a partially real set, with *scanned* (real) scores of around 300 pieces aligned to *synthesized* MIDI recordings. And (III) *RealScores_Rec*: an entirely real set, with *scanned* (real) scores of around 200 pieces and their corresponding *real audio* recordings.

As a baseline (BL), we implement the method from [14] and adapt their short-snippet-voting strategy to identify and retrieve entire music recordings and printed scores so it can operate with passages. [4] In essence, short snippets are sequentially cut out from a passage query and embedded, and are compared to all embedded snippets which were selected from passages in a search dataset of the counterpart modality, resulting in a ranked list based on the cosine distance for each passage snippet. Then the individual ranked lists are combined into a single ranking, in which the passage with most similar snippets is retrieved as the best match.

Additionally, we investigate whether our models can benefit from pre-trained cross-modal embeddings. Since both CNN encoders of our proposed network architecture (see Figure 2) are the same as in [14], we re-designed the baseline cross-modal network to accommodate our snippet dimensions (160×180 and 92×20, for sheet and audio, respectively) and trained a short-snippet embedding model also with the MSMD, as a pre-training step, and then loaded the two CNN encoders of our recurrent network with their respective pre-trained weights before training. Our hypothesis is that, by initializing the CNN encoders with parameters that were optimized to project short pairs of matching audio–sheet snippets close together onto a common latent space, models with better embedding capacity can be obtained. After loading the two CNNs with pre-trained weights, we can either freeze (FZ) them during training or just fine-tune (FT) on them. Therefore, in our experiments, we refer to these modifications of our proposed vanilla recurrent network (RNN) as RNN-FZ and RNN-FT, respectively.

Moreover, an additional CCA (canonical correlation analysis) layer [28] is used in [14] to increase the correlation of corresponding pairs in the embedding space. This CCA layer is refined in a post-training step, and we investigate whether this refinement process is beneficial to our network. In our experiments we refer to models that were initialized with pre-trained parameters from networks that had their CCA layer refined as RNN-FZ-CCA and RNN-FT-CCA.

Table 2 presents the results for all data configurations and models defined previously. To keep our experiments consistent and the comparison fair, we randomly select 534 passage pairs from sets (II) and (III) to create the retrieval scenario for their respective experiments.

An evident observation from the table is the considerable performance drop as we transition from synthetic to real music data. For all the models, the MRR drops at least

[4] The reasons we did not use the attention-based method from [17] as a baseline comparison are twofold. First we intend to compare the exact original snippet embedding architecture with and without a recurrent encoder, and adding the attention mechanism to a baseline model would introduce a significant number of additional trainable parameters, making the comparison unfair. Second, the purpose of the attention model is to compensate the musical content discrepancy between audio and sheet snippets, which is not the case for musical passages as defined here: pairs of audio–sheet music passages comprise the exact musical content (that is the reason why fragments are not fixed in time).

Table 2: Results of audio–sheet music passage retrieval, performed in both search directions, and evaluated in three types of data: (I) fully synthetic, (II) partially real and (III) entirely real. Boldfaced rows represent the best performing model per dataset.

	Audio-to-Score (A2S)					**Score-to-Audio (S2A)**				
	R@1	**R@10**	**R@25**	**MRR**	**MR**	**R@1**	**R@10**	**R@25**	**MRR**	**MR**
I MSMD (Fully synthetic)										
BL	47.56	81.68	90.80	0.592	1	51.37	83.51	92.59	0.628	1
RNN	51.12	84.46	92.88	0.627	1	54.30	85.95	94.94	0.670	1
RNN-FT	55.27	87.98	95.02	0.651	1	56.32	87.12	96.44	0.697	1
RNN-FT-CCA	**60.04**	**89.66**	**97.73**	**0.692**	**1**	**62.11**	**91.44**	**98.41**	**0.734**	**1**
RNN-FZ	50.76	84.20	92.11	0.619	1	52.90	85.21	94.12	0.658	1
RNN-FZ-CCA	52.67	86.46	92.88	0.635	1	55.67	86.30	95.34	0.682	1
II RealScores_Synth (Sheet music scans and synthetic recordings)										
BL	20.19	55.47	74.99	0.343	7	25.15	70.27	83.11	0.391	5
RNN	25.09	61.24	78.27	0.374	5	30.15	72.47	86.89	0.439	3
RNN-FT	28.87	66.41	81.32	0.447	4	33.98	75.47	88.51	0.462	2
RNN-FT-CCA	**33.36**	**69.49**	**83.88**	**0.481**	**3**	**37.35**	**79.22**	**89.95**	**0.538**	**1**
RNN-FZ	25.83	62.02	79.74	0.376	5	31.45	74.87	87.26	0.442	3
RNN-FZ-CCA	26.82	63.33	80.19	0.391	5	33.55	75.71	88.79	0.467	2
III RealScores_Rec (Sheet music scans and real recordings)										
BL	15.67	31.46	48.12	0.226	29	18.30	36.71	54.94	0.266	18
RNN	19.11	35.98	53.65	0.278	21	22.76	39.95	57.47	0.303	15
RNN-FT	22.39	39.53	57.19	0.338	18	26.76	42.77	59.38	0.371	7
RNN-FT-CCA	**26.62**	**44.81**	**60.01**	**0.362**	**7**	**29.84**	**46.71**	**60.88**	**0.435**	**4**
RNN-FZ	17.65	33.12	52.98	0.252	22	19.13	37.51	55.57	0.277	17
RNN-FZ-CCA	18.38	35.81	54.51	0.279	21	22.30	38.95	58.82	0.285	16

0.2 points to a partially real test set, and drops more than 0.3 points when moving to the entirely real data. Moreover, as mentioned in Subsection 3.1, the passage retrieval metrics of the S2A direction are better than those of A2S for all models and scenarios.

Our recurrent model RNN and its variants outperform the baseline approach in all retrieval scenarios for all evaluation metrics. In our findings, we did not see noticeable improvements when the pre-loaded encoders were frozen during training. In fact, for some configurations (scenarios I and III) the evaluation metrics were slightly worse than those from the vanilla RNN model. When the CNN encoders are pre-loaded and enabled for fine-tuning, we observe the largest improvements over RNN and subsequently over BL. Moreover, the models initialized with pre-trained weights from CCA-refined networks (RNN-FT-CCA) achieved the best overall results, for all test datasets and search directions.

In addition to the overall absolute improvements, we observe that the performance drop between synthetic and real datasets shrinks with our proposed models, specially with RNN-FT-CCA. In comparison with the baseline, the I-to-III MRR gap is reduced by 0.036 and 0.06 points in the directions A2S and S2A, respectively.

The results we obtained and summarized in Table 2 indicate that introducing a recurrent layer to learn longer contexts of musical content is beneficial in our cross-modal retrieval problem. However the real-data generalization problem is still evident, and in Section 4 we discuss potential solutions to address such issues.

3.3 Experiment 3: Global tempo variations

In this experiment, we investigate the robustness of our system to global tempo changes. To this end, the pieces of the MSMD test dataset are re-rendered with different tempo ratios $\rho \in \{0.5, 0.66, 1, 1.33, 2\}$ ($\rho = 0.5$ means the tempo was halved and $\rho = 2$ stands for doubling the original tempo). A similar study was conducted in [17] for retrieval of short audio–sheet snippets.

Table 3 summarizes the MRR values obtained for each tempo re-rendering, where the baseline method is compared with our proposed recurrent model. We notice the general trend that the MRR gets worse as the tempo ratio is farther from $\rho = 1$ (original tempo). This behavior is somehow expected because the new tempo renditions are more extreme than the tempo changes the model has seen during training.

Besides the better MRR values of the proposed network, an important improvement concerns the performance drop when changing from $\rho = 1$ to $\rho = 0.5$ (slower renditions). The MRR gap between these tempo ratios drops from 0.12 to 0.1 and from 0.09 to 0.07 points for the A2S and S2A directions, respectively, when comparing our net-

Model	$\rho = 0.5$	$\rho = 0.66$	$\rho = 1$	$\rho = 1.33$	$\rho = 2$
BL	0.47	0.54	0.59	0.52	0.40
RNN	0.53	0.59	0.63	0.58	0.43

(a) A2S search direction.

Model	$\rho = 0.5$	$\rho = 0.66$	$\rho = 1$	$\rho = 1.33$	$\rho = 2$
BL	0.54	0.59	0.63	0.56	0.48
RNN	0.60	0.64	0.67	0.61	0.50

(b) S2A search direction.

Table 3: MRR for different tempo renderings of the test pieces of MSMD in both (a) audio-to-sheet and (b) sheet-to-audio retrieval directions. We evaluate both baseline and RNN models.

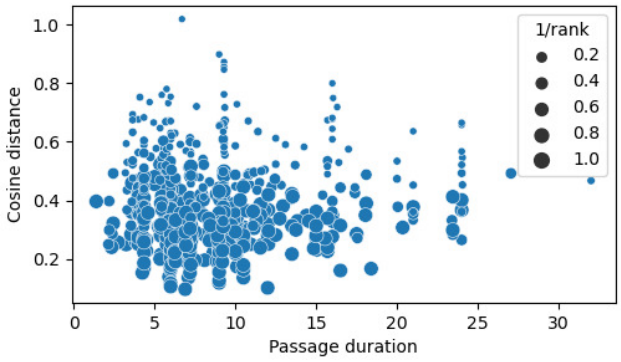


Figure 4: Cosine distance in the embedding space in relation to the respective audio passage duration of 534 pairs from the MSMD test set. The cosine distances were computed with the RNN model.

work with the baseline. This indicates that the recurrent model is more robust to global tempo variations and can operate well with longer audio passages.

3.4 Experiment 4: Qualitative analysis

To get a better understanding of the behavior of our proposed network, in this last experiment we take a closer look at the shared embedding space properties. Figure 4 shows the distribution of the pairwise cosine distances between the passage pairs from the MSMD test set, in relation to the duration (in seconds) of their respective audio passages. Moreover, we scale the point sizes in the plot so they are proportional to their individual precision values (inverse of the rank values), when considering the S2A experimental setup.

An interesting behavior in this visualization is the size of the points increasing as the cosine distance decreases. It is expected that passage pairs with smaller distances between them, meaning that they are closer together in the embedding space, would be lead to better retrieval ranks.

Another interesting aspect of this distribution concerns the proportion of larger cosine distances as the audio duration of the passages increases. For example, between five and ten seconds, there are more large points observed than smaller ones, while between 20 and 25 seconds, the proportion is roughly equal. This indicates that, in our test set, embeddings from shorter passages of audio are still located closer to their sheet counterparts in comparison with longer audio passages, despite our efforts to design a recurrent networks that learns from longer temporal contexts.

4. CONCLUSION AND FUTURE WORK

We have presented a novel cross-modal recurrent network for learning correspondences between audio and sheet music passages. Besides requiring only weakly-aligned music data for training, this approach overcomes the problems of intrinsic global and local tempo mismatches of previous works that operate on short and fixed-size fragments. Our proposed models were validated in a series of experiments under different retrieval scenarios and generated better results when comparing with baseline methods, for all possible configurations.

On the other hand, a serious generalization gap to real music data was observed, which points us to the next stages of our research. A natural step towards making deep-learning-based cross-modal audio–sheet music retrieval more robust would be to include real and diverse data that can be used for training models. However such data with suitable annotations are scarce, and recent advances in end-to-end full-page optical music recognition [29] can be a possible solution to learn correspondences on the score page level. Moreover, the powerful transformers [30] are potential architectures to learn correspondences from even longer audio recordings, accommodating typical structural differences between audio and sheet music, such as jumps and repetitions.

5. ACKNOWLEDGMENTS

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